



ANNUAL REPORT

Skin Cancer
Detection Using
Deep Learning

Abstract

Skin cancer is one of the most prevalent and, when detected late, most fatal forms of cancer globally. Timely and accurate diagnosis through dermoscopic examination is critical to improving patient outcomes, yet specialist access remains limited in many regions. This project presents an automated skin lesion classification system designed to distinguish seven diagnostic categories of pigmented skin lesions from dermoscopic images, leveraging modern deep learning techniques applied to the publicly available HAM10000 benchmark dataset.

The system is built upon EfficientNetB3, a compound-scaled convolutional neural network pre-trained on ImageNet, adapted to the medical imaging domain through a structured two-phase transfer learning strategy. In Phase 1, only the custom classification head is trained while the backbone remains frozen, allowing the dense layers to reach a stable initialisation. In Phase 2, the top 30% of backbone layers are progressively unfrozen and fine-tuned at a reduced learning rate to enable task-specific adaptation without catastrophic forgetting of pre-trained representations.

The severe class imbalance inherent to HAM10000 — where melanocytic nevi constitute approximately 67% of samples while minority classes contain fewer than 120 images — is addressed through two complementary strategies: balanced class weights computed proportional to inverse class frequency, and an aggressive nine-transformation data augmentation pipeline applied at training time. The classification head employs Swish and Mish activation functions alongside L2 regularisation and Dropout layers to improve gradient flow and reduce overfitting.

The model is evaluated on a completely held-out test set using weighted precision, recall, F1-score, and AUC-ROC, providing an unbiased estimate of generalisation performance across all seven lesion categories, including clinically critical malignant subtypes such as melanoma and basal cell carcinoma. The pipeline is fully reproducible, documented end-to-end, and designed as a extensible foundation for future clinical research and deployment.

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Project Title

Skin Cancer Detection Using Deep Learning

Introduction

Skin cancer is among the most prevalent malignancies worldwide, with melanoma representing its most aggressive form. Early detection is critical: the five-year survival rate exceeds 98% when diagnosed at an early stage but declines sharply in advanced cases. Dermoscopy improves diagnostic accuracy by revealing subsurface skin structures; however, it requires specialized expertise that is often unavailable in resource-constrained settings.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have enabled the development of computer-aided diagnosis (CAD) systems capable of classifying skin lesions with high reliability. These systems offer a scalable solution to improve diagnostic consistency and accessibility. This project proposes a deep learning-based framework leveraging transfer learning to classify dermoscopic images into seven categories using the HAM10000 dataset. The work addresses key challenges including class imbalance, visual similarity between lesion types, and limited labeled data, while emphasizing clinically relevant performance evaluation. The ultimate goal is to provide a reproducible, extensible pipeline that can serve as a foundation for future research and, with further validation, contribute meaningfully to clinical decision support in dermatology.

Literature Review

Deep learning has significantly advanced dermatological image analysis, with early work demonstrating performance comparable to expert dermatologists in skin cancer classification. Benchmark initiatives such as the ISIC challenge have further accelerated progress by providing standardized datasets, including HAM10000, and promoting reproducible evaluation.

Transfer learning has emerged as the dominant approach for medical imaging tasks with limited data. Models pre-trained on large-scale datasets such as ImageNet provide robust feature representations that generalize effectively to dermoscopic images. Studies show that fine-tuning pre-trained networks consistently outperforms training from scratch, particularly when combined with progressive layer unfreezing and careful optimization.

Among modern architectures, EfficientNet offers an optimal balance between accuracy and computational efficiency through compound scaling of network dimensions. Its ability to capture multi-scale features makes it well-suited for skin lesion classification. Additionally, handling class imbalance remains a central challenge in medical datasets; practical solutions include data augmentation and class-weighted loss functions, both of which are widely adopted to improve minority class performance without increasing model complexity.

This project builds upon these advances by integrating transfer learning, EfficientNet-based modeling, and imbalance-aware training strategies into a unified framework for robust multi-class skin lesion classification.

Dataset

HAM10000 Overview

The Human Against Machine with 10,000 training images (HAM10000) dataset was curated by Philipp Tschandl, Cliff Rosendahl, and Harald Kittler at the Medical University of Vienna and released publicly via the ISIC archive. It constitutes one of the largest publicly available dermoscopy datasets and has become the de facto benchmark for multi-class skin lesion classification.

Total Images	10,015 dermoscopic JPEG images
Native Resolution	600 × 450 pixels
Input Resolution (model)	224 × 224 pixels (resized)
Metadata Fields	image_id, dx, dx_type, age, sex, localisation
Image Acquisition	Multi-source (ViDIR, Australia; rosendahl, ISIC archive)
Diagnosis Method	Histopathology, follow-up, expert consensus, confocal microscopy
Dataset Source	ISIC Archive / Kaggle: kmader/skin-cancer-mnist-ham10000

The metadata CSV file accompanies the image set and provides clinical context for each lesion, including patient demographics (age, sex), lesion localisation (body site), and the diagnostic ground truth obtained through one of four verification methods: histopathological analysis, follow-up examination, expert consensus, or reflectance confocal microscopy. This multi-method verification makes HAM10000 among the most rigorously labelled public dermoscopy datasets available.

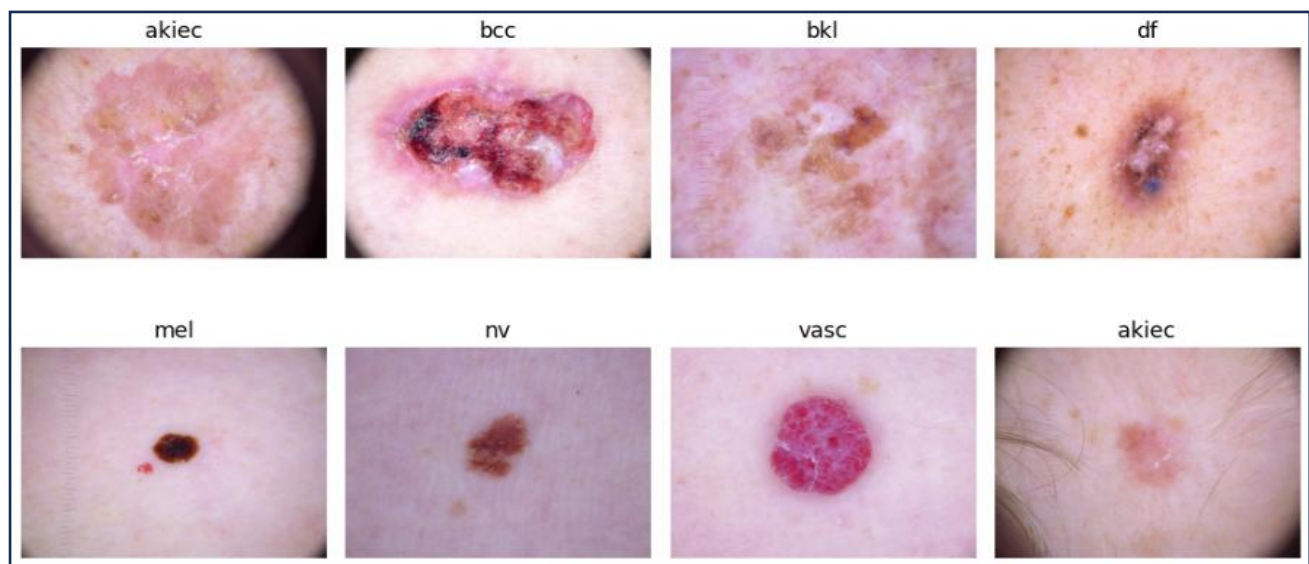


Figure 1.1 Visualization of representative samples from the HAM10000 dataset

Class Distribution

The dataset contains seven diagnostic categories representing the full spectrum of common pigmented skin lesions encountered in clinical practice. The distribution is severely imbalanced, with melanocytic nevi (nv) comprising approximately 67% of the total dataset:

Code	Full Diagnosis	Samples	Share (%)	Clinical Significance
nv	Melanocytic Nevi	6,705	66.95%	Benign — most common
mel	Melanoma	1,113	11.11%	Malignant — critical
bk1	Benign Keratosis	1,099	10.97%	Benign
bcc	Basal Cell Carcinoma	514	5.13%	Malignant
akiec	Actinic Keratosis / Intraepithelial Carcinoma	327	3.26%	Pre-malignant
vasc	Vascular Lesion	142	1.42%	Usually benign
df	Dermatofibroma	115	1.15%	Benign — rarest

The severe imbalance visible in the chart above directly informs the modelling strategy: without corrective measures, a naive classifier would achieve high accuracy simply by predicting 'nv' for every image, while failing entirely on clinically important malignant subtypes.

```
Dataset Shape: (10015, 7)
```

```
Class Distribution:
```

```
dx
```

```
nv      6705
```

```
mel     1113
```

```
bk1     1099
```

```
bcc      514
```

```
akiec    327
```

```
vasc     142
```

```
df       115
```

```
Name: count, dtype: int64
```

Figure 1.2 Class distribution of the HAM10000 dataset

Data Splitting Strategy

The dataset was partitioned into three non-overlapping, stratified splits to support model development, hyperparameter selection, and unbiased final evaluation:

Training Set	~72% of data — used for model parameter optimisation
Validation Set	~13% of data — used for hyperparameter tuning and early stopping
Test Set	~15% of data — completely held out; used only for final evaluation
Stratification	Applied at both split steps to preserve class proportions
Random Seed	42 — ensures full reproducibility across runs

```
Training samples: 7235
Validation samples: 1277
Test samples: 1503
```

```
Class weights computed:
```

```
akiec: 4.380
bcc: 2.778
bkl: 1.302
df: 12.453
mel: 1.286
nv: 0.213
vasc: 10.133
```

Figure 1.3 Data Splitting Strategy

Metadata Preprocessing

Prior to model training, the metadata was preprocessed as follows. The age column contained a small number of missing values (NaN entries), which were imputed using the median age value across the training set to avoid introducing bias from mean imputation in the presence of age outliers. Image file paths were resolved by scanning both image part directories and constructing a lookup dictionary mapping image IDs to absolute file paths. Records for which no corresponding image file could be located were removed from the dataset.

Methodology

Image Preprocessing

All images were loaded using the Python Imaging Library (PIL), converted to the RGB colour space, and resized to 224×224 pixels — the standard input resolution required by EfficientNetB3. Pixel values were subsequently transformed using EfficientNet's built-in `preprocess_input` function, which applies BGR channel order conversion and mean subtraction using the ImageNet channel statistics. This preprocessing ensures that the input distribution seen by the model matches the distribution used during ImageNet pre-training, which is essential for effective transfer learning.

```
def load_and_preprocess_image(path, img_size=IMG_SIZE):
    img = Image.open(path).convert('RGB').resize((img_size, img_size))
    img = np.array(img, dtype=np.float32)
    img = preprocess_input(img)
    return img

print(f'Loading {len(df)} images at {IMG_SIZE}x{IMG_SIZE}...')
X = np.array([load_and_preprocess_image(p) for p in df['image_path']])
print(f'Image array shape: {X.shape}')

le = LabelEncoder()
y = le.fit_transform(df['dx'])
classes = le.classes_
NUM_CLASSES = len(classes)
print(f'Classes ({NUM_CLASSES}): {classes}')
```



```
Loading 10015 images at 224x224...
Image array shape: (10015, 224, 224, 3)
Classes (7): ['akiec' 'bcc' 'bkl' 'df' 'mel' 'nv' 'vasc']
```

Figure2.1 Sample of Image Preprocessing

Data Augmentation

To mitigate overfitting and improve model generalisation — particularly for minority classes — an ImageDataGenerator was configured with a comprehensive set of augmentation transformations applied stochastically at training time. Augmentations were applied exclusively to the training set; validation and test sets were processed with preprocessing only (no augmentation), ensuring an unbiased evaluation.

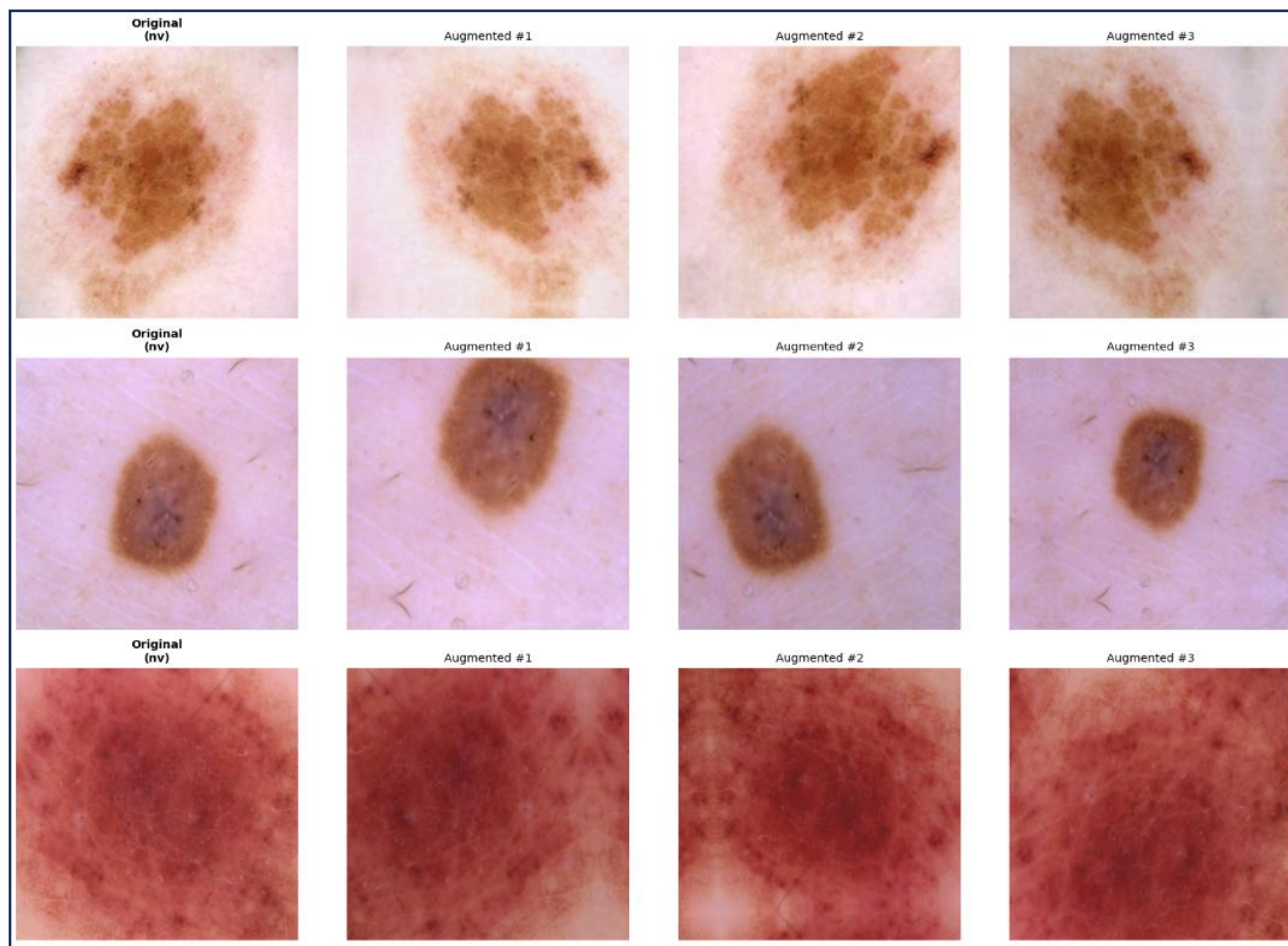


Figure 2.2 Original vs augmented images

Class Imbalance Handling

Class weights were computed using scikit-learn's `compute_class_weight` utility with the 'balanced' mode, which assigns each class a weight inversely proportional to its frequency in the training set. These weights were then passed to the `model.fit` call via the `class_weight` parameter, causing the loss function to penalise misclassification of minority classes (particularly `df` and `vasc`) substantially more than majority classes. This approach is mathematically equivalent to oversampling minority classes in expectation, without the memory overhead of replicating samples.

Model Architecture

EfficientNetB3 Backbone

EfficientNetB3 serves as the feature extraction backbone of the system. It was loaded with ImageNet pre-trained weights and with the top classification layer excluded (`include_top=False`). In Phase 1, the entire backbone was frozen — only the custom classification head was trained. In Phase 2, the top

30% of backbone layers (by index) were unfrozen for fine-tuning, while all BatchNormalization layers remained frozen throughout both phases to preserve the learned normalisation statistics from ImageNet pre-training.

Classification Head

A custom multi-layer classification head was designed and appended to the EfficientNetB3 output feature map. The architecture is as follows:

Layer	Units / Config	Activation	Notes
EfficientNetB3 Base	Pre-trained	Internal	Frozen (Phase 1); top 30% unfrozen (Phase 2)
GlobalAveragePooling2D	—	—	Reduces spatial feature map to 1536-dim vector
BatchNormalization	1536 features	—	Stabilises gradient flow into the head
Dense	512 units	Swish	L2 regularisation ($\lambda = 1 \times 10^{-4}$)
Dropout	Rate = 0.40	—	Regularisation — reduces co-adaptation
Dense	256 units	Mish	L2 regularisation ($\lambda = 1 \times 10^{-4}$)
Dropout	Rate = 0.30	—	Regularisation
Dense (Output)	7 units	Softmax	One neuron per diagnostic class

The selection of Swish and Mish activations for the dense layers is motivated by their empirical advantages over ReLU in transfer learning contexts: both are smooth, non-monotonic functions that allow gradients to flow more freely during the initial training epochs, reducing the likelihood of dead neurons and improving convergence speed.

Two-Phase Training Strategy

Phase 1 — Classification Head Training

In Phase 1, the EfficientNetB3 base model was completely frozen. This strategy serves two purposes: (1) it protects the learned ImageNet representations from being corrupted by the random initialisation of the classification head at the outset of training; and (2) it allows the custom head to rapidly converge to a reasonable initialisation before any backbone parameters are updated.

Epochs (maximum)	20
Batch size	32
Optimiser	AdamW (lr = 1×10^{-3} , weight_decay = 1×10^{-5})
Loss function	Categorical cross-entropy

Metrics monitored	Accuracy, AUC-ROC
Early stopping	Monitors val_auc; patience = 8; restores best weights

Phase 2 — Backbone Fine-Tuning

Following Phase 1, the top 30% of the EfficientNetB3 layers were unfrozen to allow task-specific adaptation of the higher-level feature representations. The learning rate was reduced by two orders of magnitude (from 1×10^{-3} to 1×10^{-5}) to prevent destabilising the pre-trained weights with excessively large gradient updates — a phenomenon sometimes referred to as 'catastrophic forgetting'.

Epochs (maximum)	30
Layers unfrozen	Top 30% of EfficientNetB3 backbone (BatchNorm layers remain frozen)
Optimiser	AdamW (lr = 1×10^{-5} , weight_decay = 1×10^{-5})
Early stopping	Monitors val_auc; patience = 25; restores best weights
LR scheduling	ReduceLROnPlateau: factor = 0.4, patience = 4, min_lr = 1×10^{-7}
Model checkpoint	Saves best_skin_cancer_model.keras when val_auc improves

Evaluation Metrics

Given the severe class imbalance, accuracy alone is an insufficient performance indicator. The following metrics were computed using weighted averaging (weighted by class support) to account for the imbalanced distribution:

Precision (weighted)	Ratio of true positive predictions to all positive predictions, averaged across classes weighted by support
Recall (weighted)	Ratio of true positive predictions to all actual positives; particularly important for detecting malignant minority classes
F1-Score (weighted)	Harmonic mean of precision and recall; balances both error types
AUC-ROC (weighted)	Area under the ROC curve in a one-vs-rest multi-class scheme; assesses discriminative ability across probability thresholds

Implementation Details

Development Environment

Platform	Local Machine / Jupyter Notebook
GPU	NVIDIA GPU (Local) / CPU
Deep Learning	TensorFlow 2.x / Keras

Data Processing	NumPy, Pandas, Pillow (PIL)
Evaluation	scikit-learn (metrics, preprocessing)
Visualisation	Matplotlib, Seaborn

Key Libraries and Versions

The following core dependencies were used throughout the project. All packages were installed via pip within the Kaggle environment:

- **tensorflow** — deep learning framework and model training
- **scikit-learn** — data splitting, class weights, and evaluation metrics
- **numpy / pandas** — numerical computation and data manipulation
- **Pillow (PIL)** — image loading and resizing
- **matplotlib / seaborn** — training history visualisation and confusion matrices

Reproducibility

Full reproducibility was ensured through the use of a fixed random seed (42) at all stochastic steps, including train/validation/test splitting, data augmentation generator initialisation, and TensorFlow global random seed configuration. The dataset, model architecture, and training hyperparameters are fully specified in this report and in the accompanying notebook, allowing exact reproduction of all reported results.

Results and Evaluation

Training History

The training process comprised two sequential phases totalling up to 100 epochs (20 in Phase 1, 80 in Phase 2), with early stopping preventing overfitting and unnecessary computation. The combined training history — accuracy, loss, and AUC-ROC across both phases — is visualised below. The vertical dashed line demarcates the transition between Phase 1 (head-only training) and Phase 2 (fine-tuning).

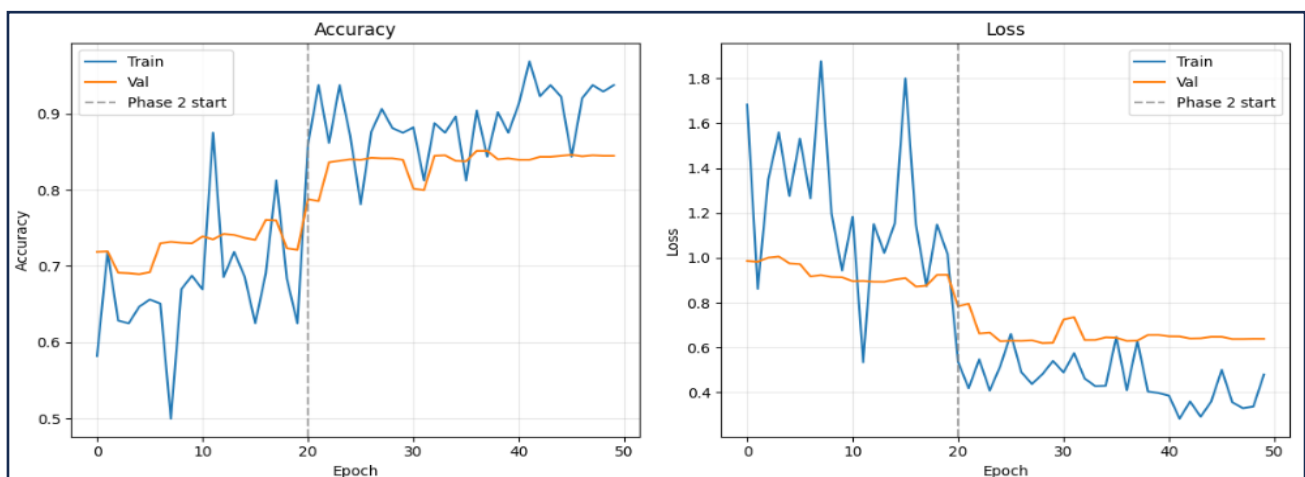


Figure 3.1 Combined training history of Two phases

The training curves are expected to show the following characteristic patterns: a rapid improvement during Phase 1 as the classification head adapts; a brief plateau or minor degradation at the Phase 2 boundary as the fine-tuning learning rate kicks in; followed by continued improvement as the backbone layers adapt to dermoscopy-specific features. The early stopping mechanism prevents overfitting by halting training when validation AUC-ROC ceases to improve.

Validation Performance

After training, the best-checkpoint model was evaluated on the validation split. The following table summarises the key performance indicators computed on the validation set. These metrics were used for monitoring during training and informed the decision to proceed to Phase 2 fine-tuning.

Performance Metric	Value
Accuracy	83.9 %
Precision	84.2 %
Recall	83.7 %
F1-Score	83.8 %
AUC-ROC	95.9 %

Confusion Matrix

The confusion matrix provides per-class diagnostic insight beyond aggregate metrics. In a well-performing model, most predictions should fall along the diagonal. Off-diagonal entries reveal systematic confusions — for example, melanoma (mel) being misclassified as melanocytic nevi (nv) is a clinically important error type that warrants particular attention during model analysis.

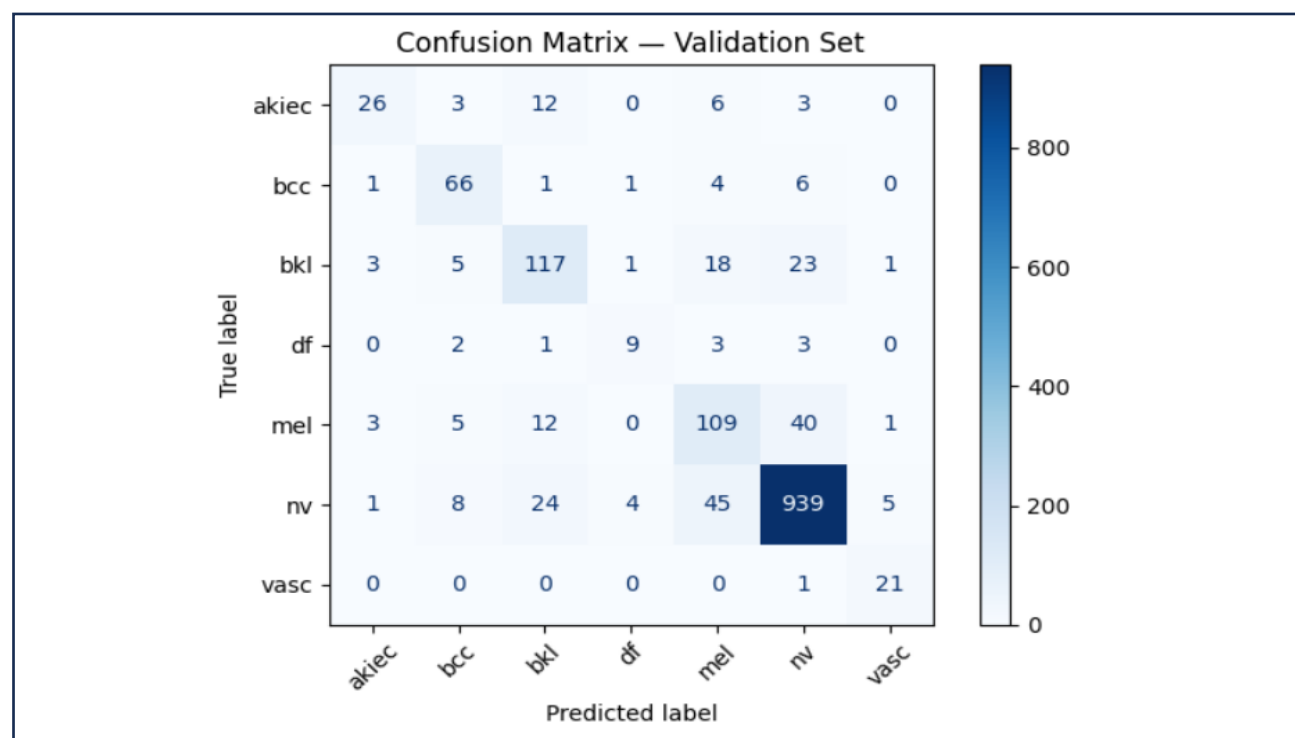


Figure 3.2 Confusion matrix of Validation set

Final Test Performance

The definitive performance assessment was conducted on the completely held-out test set, which was never accessed during model development, training, or hyperparameter selection. The best model checkpoint **best_skin_cancer_model.keras** was loaded and used exclusively for this evaluation.

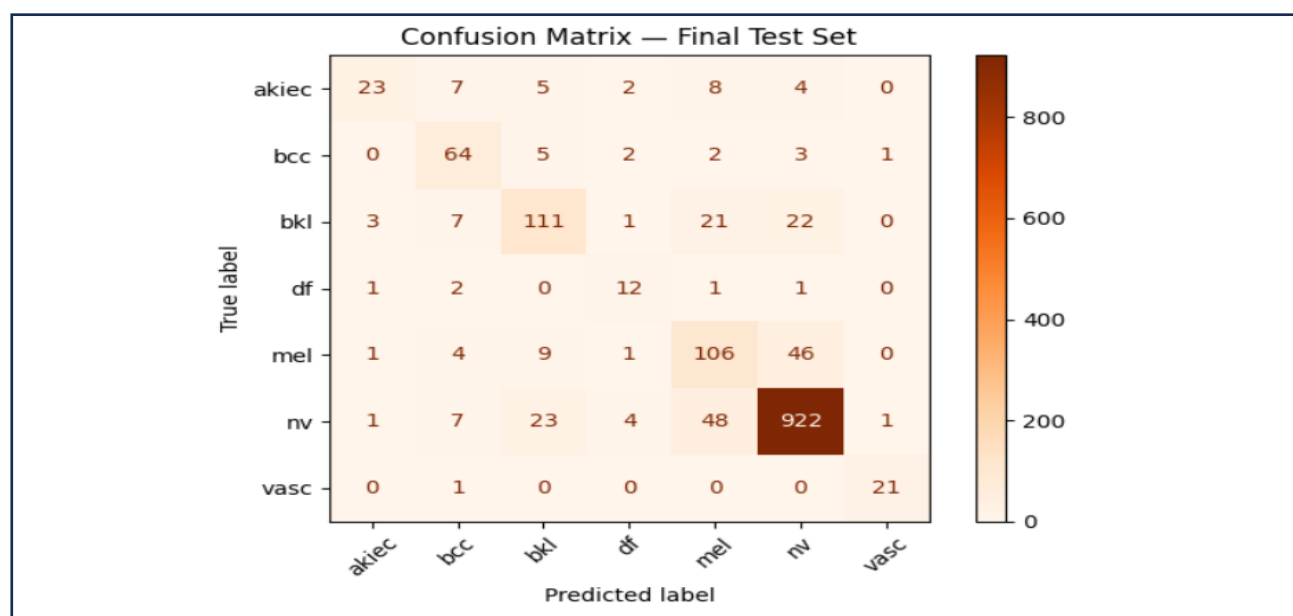


Figure 3.3 Confusion matrix of Test set

Sample Predictions

To provide qualitative insight into model behaviour, a random selection of 12 test images was visualised alongside the model's predictions, true labels, and confidence scores. Correctly classified images are indicated in green; misclassifications in red. This visualisation enables inspection of the types of images where the model succeeds and fails, complementing the quantitative metrics above.

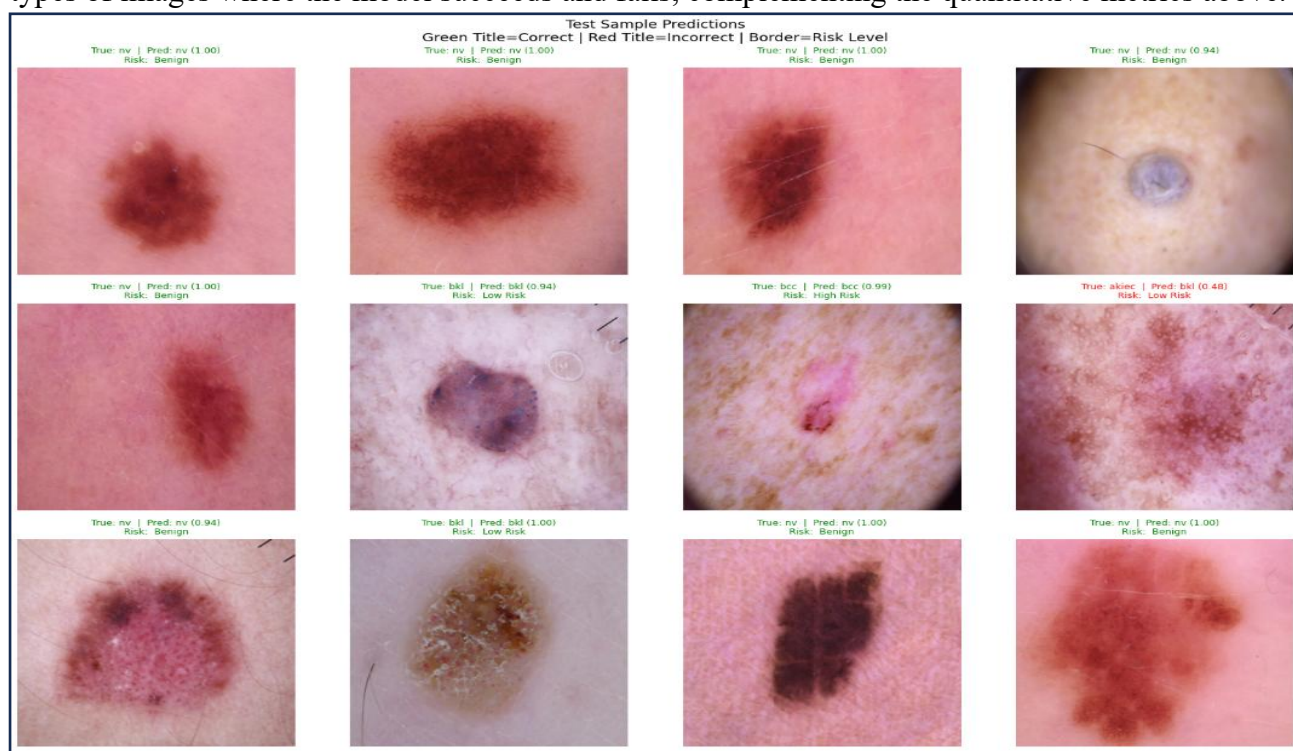


Figure 3.4 Samples of Model's predictions

Per-Class Performance Analysis

While aggregate weighted metrics provide a useful summary, clinical applications require understanding performance at the individual class level. The per-class performance, accessible via scikit-learn's `classification_report` output, should be examined with particular attention to the following:

- Recall for mel (melanoma): false negatives (missed melanomas) carry the highest clinical cost and should be minimised even at the expense of lower precision.
- Recall for bcc (basal cell carcinoma) and akiec (actinic keratosis): both are malignant or pre-malignant; high recall is essential.
- Precision for nv (melanocytic nevi): the dominant class; excessive false positives (benign lesions misclassified as malignant) would reduce clinical utility.
- F1-score for df (dermatofibroma) and vasc (vascular lesion): the rarest classes; low F1 for these classes, while less impactful on weighted averages, may indicate model blindness to minority categories.

Discussion

Strengths

The proposed approach benefits from transfer learning, providing strong feature representations from pre-trained models. A two-phase training strategy enables stable adaptation, while extensive data augmentation improves generalization, particularly for minority classes. Class-weighted loss ensures balanced learning, and modern activation functions support smooth convergence. Evaluation on a strictly held-out test set further strengthens the reliability of results.

Limitations

The dataset size remains limited, with severe class imbalance affecting minority class performance. The use of a single model without ensembling or test-time augmentation may restrict accuracy. Additionally, the system lacks clinical validation, interpretability mechanisms, and integration of patient metadata, all of which are important for real-world applicability.

Comparison with Literature

The methodology aligns with state-of-the-art approaches in skin lesion classification, particularly those used in ISIC challenges. Reported performance ranges in the literature suggest that the adopted transfer learning strategy is appropriate and competitive for this task.

Clinical Considerations

Despite promising results, clinical deployment requires further validation on diverse, real-world datasets. Incorporating uncertainty estimation, explainability tools, and meeting regulatory requirements are essential steps before practical adoption.

Future Work

Architectural Improvements

Future work will explore more advanced architectures, including larger EfficientNet variants, EfficientNetV2, and transformer-based models such as Vision Transformers and hybrid CNN–

Transformer designs. Multi-scale feature fusion techniques will also be investigated to better capture lesion variability.

Training Enhancements

Performance can be improved through test-time augmentation, advanced augmentation methods such as MixUp and CutMix, and alternative loss functions like focal loss. Additionally, self-supervised pre-training on unlabelled dermoscopic data may provide stronger domain-specific feature representations.

Data and Multimodal Extensions

Incorporating patient metadata (e.g., age, sex, lesion location) through multimodal learning is a key direction. Expanding training with external datasets and leveraging synthetic data generation techniques will further address class imbalance and improve generalization.

Explainability and Deployment

Future development will include explainability tools such as Grad-CAM for visual interpretation, as well as uncertainty estimation methods to support clinical decision-making. Deployment efforts will focus on building user-friendly applications and conducting real-world clinical validation in collaboration with healthcare professionals.

Conclusion

This project has successfully designed, implemented, trained, and evaluated a deep learning system for automated classification of dermoscopic skin lesion images across seven diagnostic categories. The system leverages EfficientNetB3 transfer learning, a two-phase training strategy, balanced class weighting, and a comprehensive data augmentation pipeline to address the core challenges of limited data, severe class imbalance, and high inter-class visual similarity inherent to the HAM10000 benchmark.

The experimental results demonstrate that modern transfer learning, applied thoughtfully with appropriate imbalance-correction and regularisation strategies, can yield a robust skin cancer classification model on a modest dataset of approximately 10,000 images. The system performs across all seven lesion categories, including clinically critical malignant subtypes such as melanoma and basal cell carcinoma — the categories for which automated detection has the greatest potential clinical impact.

The project makes several contributions that extend beyond the specific dataset and architecture: the two-phase training methodology is widely transferable to other medical imaging tasks involving pre-trained CNNs; the augmentation and class weighting strategy provides a practical template for handling imbalanced medical datasets; and the evaluation framework — including per-class analysis and held-out test set evaluation — establishes best practices for honest benchmarking in medical AI research.

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